8 The Chi-Square Distribution

Suppose a small robot is programmed to stop at a marked location. Each attempt ends a little to the left or right and a little ahead of or behind the target. We could describe these errors separately, but we might also want one number that summarizes how far an attempt missed its target.

Squaring the errors prevents positive and negative deviations from canceling. Adding the squared errors produces a measure of overall departure from the target. Similar calculations appear when we estimate a variance, combine measurements from different calibration runs, or decide whether a collection of errors is unusually large.

The **chi-square distribution** describes a particular kind of sum of squares. It connects normal observations to the sampling distributions of variance estimators. This chapter develops that connection and uses it to study positioning errors, combine calibration data, and plan how many measurements to collect.

8.1 From normal errors to a sum of squares

A standard normal random variable has mean 0 and variance 1. If we square it, the result is always nonnegative. Small squared values occur when the original observation is near zero. Large squared values occur when the observation is far from zero in either direction.

> **Note**
>
> χ is the Greek letter chi, pronounced "kye." The notation $Q \sim \chi^2_m$ means that the random variable Q has a chi-square distribution with m degrees of freedom. The superscript 2 belongs to the distribution's name. It does not instruct us to square Q again.
> Z_i denotes a standard normal random variable, with $Z_i \sim N(0,1)$. The expression Z_i^2 does mean to square that variable. The subscript i labels one variable in a collection.
> The quantity m is the distribution's degrees of freedom. In the sum-of-squares construction below, it counts the independent standard normal variables being squared and added.

If $Z \sim N(0,1)$, then

$$Z^2 \sim \chi^2_1.$$

More generally, if $Z_1, \ldots, Z_m$ are independent standard normal random variables, then

$$Q = \sum_{i=1}^{m} Z_i^2 = Z_1^2 + \cdots + Z_m^2 \sim \chi^2_m.$$

The independence and standardization requirements both matter. Squaring arbitrary normal observations does not automatically produce this distribution. Each observation must first have mean 0 and variance 1, and the squared contributions must come from independent variables.

8.1.1 Standardizing a normal observation

> **Note**
>
> For $X_i \sim N(\mu, \sigma^2)$, the variable
>
> $$Z_i = \frac{X_i - \mu}{\sigma}$$
>
> is a standardized deviation. Subtracting μ centers the observation at zero. Dividing by the standard deviation σ expresses the deviation in standard deviation units. The denominator is σ, while the denominator of the squared deviation is σ^2.

For independent observations from the same normal population,

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2 = \frac{\sum_{i=1}^{n} (X_i - \mu)^2}{\sigma^2} \sim \chi^2_n.$$

The raw sum of squared deviations has the squared units of the measurements. Dividing by the population variance removes those units and produces a chi-square variable.

This statement describes a sampling distribution under the population model. If σ^2 is unknown, we cannot directly calculate this standardized sum from the observations alone. We can still use its distribution to understand how an estimator behaves for any proposed true variance.

8.2 The density and its shape

> **Note**
>
> $f_Q(q)$ denotes the density of Q evaluated at a possible value q. The subscript identifies which variable the density describes. The function Γ is the gamma function introduced in Chapter 1. Here it supplies part of the normalizing constant that makes the density integrate to 1.

The density of $Q \sim \chi^2_m$ is

$$f_Q(q) = \frac{1}{2^{m/2}\Gamma(m/2)} q^{m/2-1} e^{-q/2}, \qquad q > 0.$$

The distribution assigns probability only to positive values, although values can be arbitrarily close to zero.

The sum-of-squares construction uses a positive integer m. The density itself also defines a valid distribution for any positive real m. The examples in this chapter use integer degrees of freedom because they arise from counting independent normal contributions.

For small degrees of freedom, the distribution is strongly skewed to the right. With one degree of freedom, the density increases without bound as we approach zero, but the total area under the curve is still 1. As the degrees of freedom increase, the distribution becomes less skewed and its center moves to the right.

```
possible_values <- seq(0.02, 45, length.out = 1000)
degrees_of_freedom <- c(1, 2, 6, 18)

par(mfrow = c(2, 2))

for (m in degrees_of_freedom) {
  plot(
    possible_values,
    dchisq(possible_values, df = m),
    type = "l",
    col = "steelblue",
    lwd = 2,
    xlab = "Possible value",
    ylab = "Density",
    main = paste("Degrees of freedom =", m)
  )
```

```
  abline(v = m, col = "firebrick", lty = 2)
}

par(mfrow = c(1, 1))
```

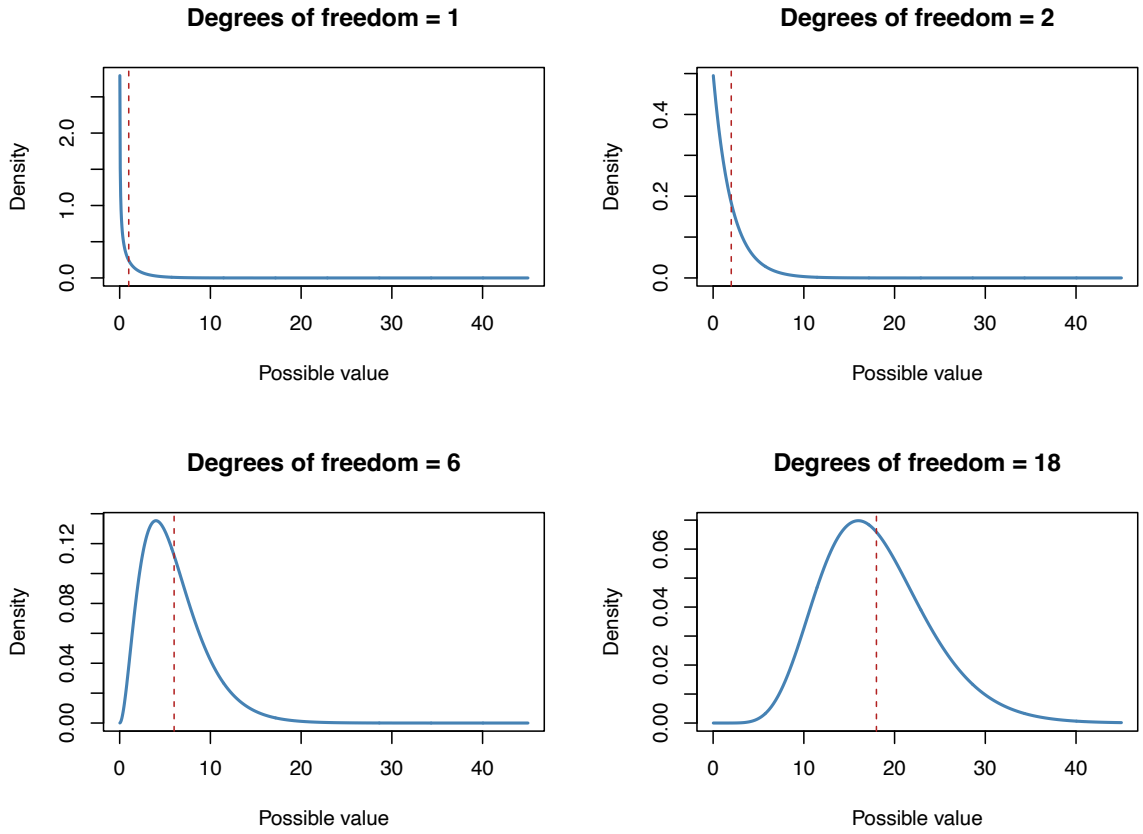


Figure 8.1: Chi-square densities for four choices of degrees of freedom. Horizontal scales are shared, while vertical scales adjust to show each shape. The curve for one degree of freedom continues upward as it approaches zero.

The dashed lines in Figure 8.1 mark the means.

8.3 A chi-square distribution is a gamma distribution

Chapter 1 introduced the gamma family using a shape parameter and a rate parameter. Its density is

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \qquad x > 0.$$

Setting $\alpha = m/2$ and $\beta = 1/2$ gives the chi-square density. Therefore,

$$\chi^2_m = \text{Gamma}\left(\frac{m}{2}, \frac{1}{2}\right),$$

where the equality means that these names describe the same probability distribution.

> **Note**
>
> In Gamma(α, β), α is the shape and β is the rate. This book uses the shape-rate convention. A gamma distribution can also be described using a scale, which is the reciprocal of the rate.
> Thus a chi-square distribution has shape $m/2$, rate $1/2$, and scale 2. A rate of $1/2$ and a scale of $1/2$ describe different distributions.

8.3.1 Mean and variance

The gamma mean and variance give

$$E[Q] = \frac{m/2}{1/2} = m$$

and

$$\text{Var}(Q) = \frac{m/2}{(1/2)^2} = 2m.$$

Consequently, the standard deviation is $\sqrt{2m}$. Both the mean and the standard deviation grow with m. However, the standard deviation relative to the mean is

$$\frac{\text{SD}(Q)}{E[Q]} = \frac{\sqrt{2m}}{m} = \sqrt{\frac{2}{m}},$$

which decreases as m grows. The distribution becomes more concentrated relative to the size of its mean.

For large integer m, the central limit theorem applied to the independent squared normal terms suggests

$$\frac{Q-m}{\sqrt{2m}} \text{ is approximately } N(0,1).$$

Here we subtract the mean and divide by the standard deviation of Q. This explains the increasingly symmetric shape without suggesting that Q itself has a fixed center or spread as m changes.

8.3.2 Multiplying by a positive constant

Suppose $Q \sim \chi^2_m$ and $c > 0$ is a fixed constant. Multiplying by c stretches the distribution horizontally:

$$cQ \sim \text{Gamma}\left(\frac{m}{2}, \frac{1}{2c}\right).$$

The gamma shape stays the same, while the rate is divided by c. The mean becomes cm and the variance becomes $2c^2m$.

The restriction $c > 0$ matters. A zero multiplier gives a constant, and a negative multiplier produces negative values rather than a gamma distribution. Also, cQ generally has a gamma distribution rather than another chi-square distribution. Changing the scale does not simply change the degrees of freedom.

8.4 Adding independent chi-square variables

> **Note**
>
> Q_1 and Q_2 name two random quantities. Their degrees of freedom are m_1 and m_2. The subscripts distinguish the quantities and their parameters. They are labels rather than exponents.

If

$$Q_1 \sim \chi^2_{m_1}, \qquad Q_2 \sim \chi^2_{m_2},$$

and Q_1 and Q_2 are independent, then

$$Q_1 + Q_2 \sim \chi^2_{m_1+m_2}.$$

For integer degrees of freedom, we can understand this by collecting the squared standard normal terms from both sums. The combined sum has $m_1 + m_2$ independent contributions.

Independence is an assumption that is very important here. For example, reusing the same random observation does not create another independent contribution. If both Q_1 and Q_2 are constructed using the same Z^2, their sum is $2Z^2$. Its variance is 8, whereas a chi-square variable with two degrees of freedom has variance 4. The two distributions are different.

8.5 Example: a robot's combined positioning error

A robot's horizontal and vertical stopping errors, measured in centimeters, are modeled as

$$X \sim N(0, 0.4^2), \qquad Y \sim N(0, 0.7^2),$$

with X and Y independent. The robot is less precise in the vertical direction. A vertical error and a horizontal error of the same physical size therefore represent different departures from their usual behavior.

To compare them on the same scale, define the combined score

$$Q = \left(\frac{X}{0.4}\right)^2 + \left(\frac{Y}{0.7}\right)^2.$$

Each standardized error is standard normal, and the two errors are independent. Thus $Q \sim \chi^2_2$.

The engineering team wants to mark a region containing 95% of stopping locations under this model. This requires a cutoff with 95% of the chi-square probability to its left.

> **Note**
>
> $q_p(m)$ denotes the p quantile of a chi-square distribution with m degrees of freedom. It satisfies
>
> $$P(Q \le q_p(m)) = p, \qquad Q \sim \chi^2_m.$$
>
> The subscript p labels the cumulative probability. The argument m specifies degrees of freedom. This p is a probability used to choose a cutoff, rather than an unknown success probability being estimated.

For two degrees of freedom,

$$q_{0.95}(2) \approx 5.991.$$

The desired region is therefore

$$\left(\frac{x}{0.4}\right)^2 + \left(\frac{y}{0.7}\right)^2 \leq 5.991.$$

This describes an ellipse centered on the target. Along the horizontal axis it extends about $0.4\sqrt{5.991} = 0.979$ centimeters in either direction. Along the vertical axis it extends about $0.7\sqrt{5.991} = 1.713$ centimeters. The shape allows larger vertical errors because the vertical error distribution is more spread out.

```
qchisq(0.95, df = 2)
```

```
[1] 5.991465
```

For a particular stop with errors $x = 0.8$ and $y = 0.7$, the score is

$$q = \left(\frac{0.8}{0.4}\right)^2 + \left(\frac{0.7}{0.7}\right)^2 = 5.$$

This stop is inside the region. The probability of a score at least this large under the model is

```
pchisq(5, df = 2, lower.tail = FALSE)
```

```
[1] 0.082085
```

The result is about 0.082. Roughly 8.2% of stops would have a combined standardized error score at least 5 if the model were an accurate representation of the process.

Note that the unstandardized squared distance $X^2 + Y^2$ would not have this chi-square distribution because the two error variances differ. Standardization is what makes their squared contributions comparable.

8.6 Using chi-square tables and R

The chi-square table supplied with Lecture 8 reports **left-tail quantiles**. Its column label p means the probability to the left of the tabulated value. To find the robot's cutoff, use the row for two degrees of freedom and the column for $p = 0.95$. The entry is approximately 5.991.

For a cutoff with 5% of the probability to its right, the required left-tail probability is $1-0.05 = 0.95$. Reading the 0.05 column instead would give a cutoff near the opposite end of the distribution.

In R, the relevant functions follow the distribution naming pattern introduced in the appendix:

Function	Input and purpose
`dchisq(x, df = m)`	Evaluate the density at `x`
`pchisq(x, df = m)`	Find the probability at or below `x`
`pchisq(x, df = m, lower.tail = FALSE)`	Find the probability above `x`
`qchisq(p, df = m)`	Find the cutoff with cumulative probability `p`
`rchisq(n, df = m)`	Generate `n` random values with `m` degrees of freedom

The argument `df` stands for degrees of freedom. It is not automatically equal to the number of random values requested from `rchisq()`.

A density height is not a tail probability. Use `pchisq()` when the cutoff is known and a probability is wanted. Use `qchisq()` when the probability is known and the cutoff is wanted.

8.7 Why chi-square distributions appear in variance estimation

Consider independent observations from $N(\mu, \sigma^2)$ with known mean μ and unknown variance σ^2. The maximum likelihood estimator of the variance is

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2.$$

> **Note**
>
> $\hat{\sigma}^2$ denotes an estimator of the population variance σ^2. This is also written $\widehat{\sigma^2}$. Its observed value is obtained by substituting the recorded observations into the formula.

The notation σ^4 means $(\sigma^2)^2$, the square of the population variance. It appears when calculating the variance of a variance estimator, which has fourth-power measurement units.

Multiplying by n/σ^2 recovers the standardized sum of squares:

$$\frac{n\hat{\sigma}^2}{\sigma^2} = \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2_n.$$

Equivalently, the estimator has the distribution of a chi-square variable multiplied by σ^2/n. The gamma scaling rule gives

$$\hat{\sigma}^2 \sim \text{Gamma}\left(\frac{n}{2}, \frac{n}{2\sigma^2} \right).$$

Its mean and variance are

$$E[\hat{\sigma}^2] = \sigma^2, \qquad \text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n}.$$

Thus the variance MLE is unbiased in this known-mean setting, with standard error

$$\text{SE}(\hat{\sigma}^2) = \sigma^2 \sqrt{\frac{2}{n}}.$$

The known mean is essential to the degrees-of-freedom count here. Later we will investigate what happens when the mean is not known.

8.8 Example: combining two calibration runs

A sensor is calibrated at two reference settings. At the first setting, its measurements have a known mean of 10 units. At the second, they have a known mean of 40 units. Assume both sets of readings are normal with the same unknown variance σ^2, and all readings are independent.

The first run contains 6 readings and the second contains 18. The question is how to combine their information about the common variance. Should the two estimated variances receive equal weight, or should the longer run contribute more?

> **Note**
>
> V_A and V_B denote the variance estimates from calibration runs A and B. Their subscripts identify the runs. The random observations are $X_1, \dots, X_6$ for A and $Y_1, \dots, Y_{18}$ for B. V_P denotes the pooled estimate that uses all squared deviations together. V_E denotes an estimate that gives the two runs equal weight. Lowercase v_A, v_B, and v_P denote numerical values computed from observed data. Here V is simply a name for an estimator.

Using the known reference means,

$$V_A = \frac{1}{6}\sum_{i=1}^{6}(X_i - 10)^2, \qquad V_B = \frac{1}{18}\sum_{j=1}^{18}(Y_j - 40)^2.$$

Their standardized sums of squares satisfy

$$\frac{6V_A}{\sigma^2} \sim \chi^2_6, \qquad \frac{18V_B}{\sigma^2} \sim \chi^2_{18}.$$

The two sums are independent because the runs are independent. Adding them gives

$$\frac{6V_A + 18V_B}{\sigma^2} \sim \chi^2_{24}.$$

The pooled estimate is

$$V_P = \frac{6V_A + 18V_B}{24} = \frac{1}{4}V_A + \frac{3}{4}V_B.$$

It averages all 24 squared deviations from their respective known means. The second run receives three times the weight because it contains three times as many observations. The known-mean likelihood calculation also makes this the MLE of the common variance.

If the observed estimates are $v_A = 0.09$ and $v_B = 0.25$ squared units, then

$$v_P = \frac{6(0.09) + 18(0.25)}{24} = 0.21.$$

The sampling distribution of the pooled estimate is

$$\frac{24V_P}{\sigma^2} \sim \chi^2_{24}, \qquad V_P \sim \text{Gamma}\left(12, \frac{12}{\sigma^2}\right).$$

Consequently,

$$E[V_P] = \sigma^2, \qquad \operatorname{Var}(V_P) = \frac{\sigma^4}{12}.$$

8.8.1 What happens if we weight the runs equally?

An alternative is

$$V_E = \frac{V_A + V_B}{2}.$$

It is also unbiased because both run-specific estimates have mean σ^2. For the observed estimates above, it gives $(0.09 + 0.25)/2 = 0.17$ squared units.

However, the estimate from the shorter run has more sampling variability. Equal weighting gives that less precise estimate half the influence. Using independence,

$$\begin{aligned}
\operatorname{Var}(V_E) &= \frac{1}{4}\operatorname{Var}(V_A) + \frac{1}{4}\operatorname{Var}(V_B) \\
&= \frac{1}{4}\left(\frac{2\sigma^4}{6} + \frac{2\sigma^4}{18}\right) \\
&= \frac{\sigma^4}{9}.
\end{aligned}$$

This is larger than $\sigma^4/12$, the variance of the pooled estimate. Both rules are correct on average, but pooling gives more stable estimates under the assumed common-variance model. If the two settings actually have different variances, that common-variance interpretation would no longer apply.

8.8.2 Simulating the two estimation rules

To see the difference, suppose the true common variance is 0.16 squared units. The following simulation generates both calibration runs and calculates both estimates from each pair of runs.

```
set.seed(132)
true_variance <- 0.16
pooled_estimates <- numeric(8000)
equal_estimates <- numeric(8000)

for (i in seq_along(pooled_estimates)) {
  readings_a <- rnorm(6, mean = 10, sd = sqrt(true_variance))
```

```r
  readings_b <- rnorm(18, mean = 40, sd = sqrt(true_variance))
  estimate_a <- mean((readings_a - 10)^2)
  estimate_b <- mean((readings_b - 40)^2)
  pooled_estimates[i] <- (6 * estimate_a + 18 * estimate_b) / 24
  equal_estimates[i] <- (estimate_a + estimate_b) / 2
}

bin_edges <- seq(
  0, max(pooled_estimates, equal_estimates) + 0.01, by = 0.01
)
possible_variances <- seq(0, max(bin_edges), length.out = 500)

par(mfrow = c(1, 2))

hist(
  pooled_estimates,
  breaks = bin_edges,
  probability = TRUE,
  col = "skyblue", border = "white",
  ylim = c(0, 11),
  xlab = "Estimated variance", main = "Pool all 24 readings"
)
lines(
  possible_variances,
  dgamma(possible_variances, shape = 12, rate = 12 / true_variance),
  col = "steelblue", lwd = 2
)
abline(v = true_variance, col = "firebrick", lty = 2, lwd = 2)

hist(
  equal_estimates,
  breaks = bin_edges,
  probability = TRUE,
  col = "skyblue", border = "white",
  ylim = c(0, 11),
  xlab = "Estimated variance", main = "Weight the two runs equally"
)
abline(v = true_variance, col = "firebrick", lty = 2, lwd = 2)

par(mfrow = c(1, 1))
```

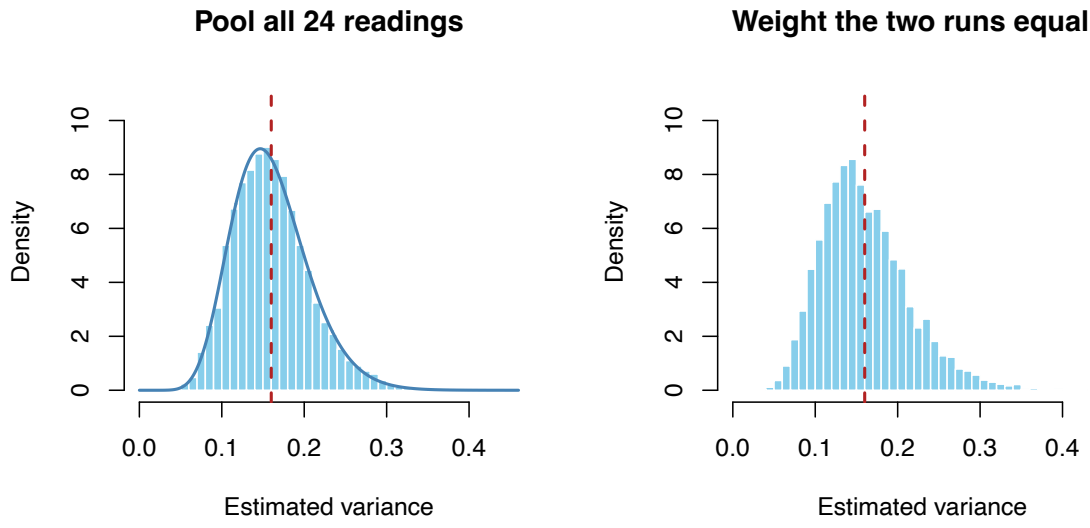


Figure 8.2: Sampling distributions of two variance estimates from calibration runs of sizes 6 and 18. Both are centered at the true variance, but the pooled estimate is less variable. The blue curve is the exact gamma density for the pooled estimate.

The panels in Figure 8.2 use the same axes and histogram bins. The exact standard errors are about 0.0462 squared units for pooling and 0.0533 squared units for equal weighting. The agreement between the left histogram and its gamma curve illustrates how normal measurements lead to a gamma sampling distribution after squaring and averaging deviations.

8.9 Example: planning the size of a calibration study

Suppose the calibration team wants its variance estimate to be within 25% of the true variance with probability at least 0.90. It can afford a total of 24, 48, or 96 independent readings. Each reading will be centered at its known reference mean, and the common normal variance model is assumed to hold.

> **Note**
>
> V_n denotes the pooled variance estimate based on a total of n readings. The ratio V_n/σ^2 expresses the estimated variance as a fraction of the true variance. A ratio of 1 is exact agreement, while 0.75 and 1.25 mark errors of 25% below and above the truth.
> The interval condition $0.75\sigma^2 \le V_n \le 1.25\sigma^2$ concerns the random estimator for a fixed true variance. It is a sampling probability used for planning.

The standardized pooled sum has distribution

$$\frac{nV_n}{\sigma^2} \sim \chi^2_n.$$

Therefore,

$$P\left(0.75\sigma^2 \leq V_n \leq 1.25\sigma^2\right) = P\left(0.75n \leq \frac{nV_n}{\sigma^2} \leq 1.25n\right).$$

The unknown variance cancels from the bounds. We can calculate the probability for each proposed sample size without choosing a numerical value of σ^2.

```
candidate_sizes <- c(24, 48, 96)
within_tolerance <- pchisq(1.25 * candidate_sizes, df = candidate_sizes) -
  pchisq(0.75 * candidate_sizes, df = candidate_sizes)

data.frame(
  sample_size = candidate_sizes,
  probability = round(within_tolerance, digits = 3)
)
```

```
  sample_size probability
1          24       0.618
2          48       0.784
3          96       0.919
```

The probabilities are approximately 0.618, 0.784, and 0.919. Of the three proposed designs, 96 readings is the smallest that meets the 90% requirement. This does not claim that 96 is the smallest possible integer sample size. It is the smallest among the three options being considered.

This calculation turns a sampling distribution into a study-design decision. More observations improve the chance of a sufficiently accurate variance estimate. The result depends on the normal model, independent readings, a common variance, and known reference means.

8.10 Chapter summary

- A chi-square variable with integer degrees of freedom is a sum of independent squared standard normal variables.
- The distribution χ^2_m is a gamma distribution with shape $m/2$ and rate $1/2$.

- Its mean is m, its variance is $2m$, and its relative spread decreases as m grows.
- Multiplying a chi-square variable by a positive constant changes the gamma rate while leaving the shape unchanged.
- Independent chi-square variables add to another chi-square variable whose degrees of freedom are the sum of the individual degrees of freedom.
- Chi-square quantiles turn a specified probability into a cutoff. The lecture table uses cumulative probability to the left of that cutoff.
- With a known normal mean, a variance MLE has a scaled chi-square, or gamma, sampling distribution.